IIT JAM Mathematics

Basic Algebra: P&C, Binomial Theorem Cheat Sheet

1. Fundamental Principles

Addition Principle (OR): If task A can be done in m ways and task B in n ways, and they cannot be done simultaneously, then A or B can be done in $m+n$ ways.
Multiplication Principle (AND): If task A can be done in m ways and task B in n ways, then doing both sequentially can be done in $m \times n$ ways.

2. Permutations (Arrangements)

Definition: Arrangement of objects in a specific order.
Factorial: $n! = n(n-1)(n-2)\dots 1$ (with $0! = 1$).
Linear Permutation (^nP_r): Number of arrangements of r objects chosen from n distinct objects:

$$^nP_r = P(n,r) = \frac{n!}{(n-r)!}$$

Permutation of Multisets: The number of arrangements of n objects where p_1 are of type 1, p_2 are of type 2, $\dots$, p_k are of type k:

$$\frac{n!}{p_1!p_2!\dots p_k!}$$

Circular Permutation:

- n distinct objects in a circle: $(n-1)!$
- If clockwise and anti-clockwise arrangements are not distinct (e.g., a necklace of beads): $\frac{(n-1)!}{2}$

3. Combinations (Selections)

Definition: Selection of objects where order does not matter.
Formula (^nC_r): Number of ways to choose r objects from n distinct objects:

$$^nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{^nP_r}{r!}$$

Key Properties:

- **Symmetry:** $\binom{n}{r} = \binom{n}{n-r}$
- **Pascal's Identity:** $\binom{n}{r} + \binom{n}{r-1} = \binom{n+1}{r}$
- **Extraction:** $\binom{n}{r} = \frac{n}{r}\binom{n-1}{r-1}$

Combinations with Repetition (Stars & Bars): Number of ways to select r objects from n distinct types (unlimited supply):

$$\binom{n+r-1}{r} = \binom{n+r-1}{n-1}$$

Application: The number of non-negative integer solutions to $x_1 + x_2 + \dots + x_n = r$ is $\binom{n+r-1}{n-1}$.
The number of positive integer solutions is $\binom{r-1}{n-1}$.

4. Advanced Counting Methods

Derangements (D_n): Number of permutations of n elements such that no element appears in its original position:

$$D_n = n!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots + (-1)^n \frac{1}{n!}\right)$$

Recursive relations:
$D_n = (n-1)(D_{n-1} + D_{n-2})$ and $D_n = nD_{n-1} + (-1)^n$.
Inclusion-Exclusion Principle:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$\begin{aligned} |A \cup B \cup C| = |A| + |B| + |C| \\ - |A \cap B| - |B \cap C| - |C \cap A| \\ + |A \cap B \cap C| \end{aligned}$$

Pigeonhole Principle: If n objects are placed into m containers and $n > m$, at least one container must hold at least $\lceil n/m \rceil$ objects.

5. Binomial Theorem

Expansion for Positive Integer n:

$$(x+y)^n = \sum_{r=0}^{n} \binom{n}{r} x^{n-r} y^r$$

General Term: $T_{r+1} = \binom{n}{r} x^{n-r} y^r$
Middle Term(s):

- If n is **even**, there is 1 middle term: $T_{\frac{n}{2}+1}$.
- If n is **odd**, there are 2 middle terms: $T_{\frac{n+1}{2}}$ and $T_{\frac{n+3}{2}}$.

Greatest Binomial Coefficient: Occurs at the middle term(s). For n even, it is $\binom{n}{n/2}$. For n odd, they are $\binom{n}{(n-1)/2} = \binom{n}{(n+1)/2}$.

6. Properties of Binomial Coefficients

Let $C_r = \binom{n}{r}$.

- **Sum of coefficients:** $\sum_{r=0}^{n} C_r = 2^n$
- **Alternating sum:** $\sum_{r=0}^{n} (-1)^r C_r = 0$
- **Sum of evens/odds:** $C_0 + C_2 + \dots = C_1 + C_3 + \dots = 2^{n-1}$
- **Sum of squares:** $\sum_{r=0}^{n} C_r^2 = \binom{2n}{n}$
- **Weighted sum:** $\sum_{r=1}^{n} rC_r = n2^{n-1}$
- **Vandermonde's Identity:** $\sum_{r=0}^{k} \binom{m}{r}\binom{n}{k-r} = \binom{m+n}{k}$

7. Extensions of Binomial Theorem

Multinomial Theorem:

$$(x_1 + x_2 + \dots + x_k)^n = \sum \frac{n!}{n_1!n_2!\dots n_k!} x_1^{n_1} x_2^{n_2} \dots x_k^{n_k}$$

where the sum is over all non-negative integers $n_1, \dots, n_k$ such that $n_1 + \dots + n_k = n$.
Note: The number of distinct terms in the expansion is $\binom{n+k-1}{k-1}$.
Binomial Theorem for any Index:
If n is negative or fractional, and $|x| < 1$, then:

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots$$

Important special cases for $|x| < 1$:

- $(1-x)^{-1} = 1 + x + x^2 + x^3 + \dots$
- $(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \dots$
- $(1-x)^{-n} = \sum_{r=0}^{\infty} \binom{n+r-1}{r} x^r$